



Analog IC Design ASSESSMENT 6

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Submitted to
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1 Objective and Specifications

The experiment involves the design and simulation of CMOS cascode amplifiers using Cadence Virtuoso. The work is divided into two parts. Part A uses a resistive load, while Part B uses an NMOS cascode stage with a PMOS cascode current-source load. The structure and presentation follow the supplied Assessment–6 reference, while the numerical values and Cadence plots in this report are replaced by the submitted simulation outputs.

Design Specifications

Table 1: Design specifications used for the cascode amplifier experiment

Parameter	Specification
Supply voltage, V_{DD}	1.8 V
Target voltage gain	20 dB
Target power consumption	1 mW
Target 3-dB bandwidth	20 MHz
Load capacitance, C_L	5 pF
Technology	UMC 180 nm CMOS

The final submitted schematics use a nominal channel length of 180 nm. The resistive-load circuit uses a 1.6 k Ω drain resistor and 5 pF load capacitor. The current-source-loaded circuit uses four transistors with individually selected widths from the Cadence width sweeps.

2 Theory and Transfer Functions

2.1 Cascode Amplifier

A cascode amplifier combines a common-source device with a common-gate device. The upper device reduces the variation of the lower transistor drain voltage and increases the small-signal output resistance. Consequently, the voltage gain can be increased without requiring an excessively large load resistor.

For a resistive-load cascode stage, a useful first-order representation is

$$A_v \approx -g_m R_{out} \quad (1)$$

where g_m is the transconductance of the input transistor and R_{out} is the effective small-signal resistance looking into the output node. For a practical cascode, R_{out} is substantially higher than the drain resistance of a single transistor because of the cascode action.

2.2 Resistive-Load Stage

With a resistor R_D and a load capacitor C_L at the output, the dominant pole is approximately

$$f_p \approx \frac{1}{2\pi R_D C_L} \quad (2)$$

For the specified 20 MHz bandwidth and 5 pF load, the corresponding first-order drain resistance is used as a starting point for the design.

2.3 PMOS Cascode Current-Source Load

In Part B, the resistor is replaced by two PMOS devices arranged as a cascode current-source load. A simplified small-signal output resistance is the parallel combination of the NMOS and PMOS cascode resistances. A compact low-frequency form is

$$A_v \approx -g_{m,n} R_{out,eq} \quad (3)$$

with

$$R_{out,eq} \approx R_{out,n} \parallel R_{out,p} \quad (4)$$

The PMOS cascode load therefore presents a high output resistance to the NMOS cascode stage.

2.4 Pole-Zero Representation

A measured amplifier response can be represented in compact form as

$$H(s) \approx A_0 \frac{1 - s/\omega_z}{(1 + s/\omega_{p1})(1 + s/\omega_{p2}) \dots} \quad (5)$$

The dominant pole controls the first major high-frequency roll-off, while higher-order poles and zeros determine the later-frequency behaviour.

2.5 Gain Crossover, Gain Margin and Phase Margin

For a feedback-loop transfer function $L(s)$, the gain-crossover frequency is defined by

$$|L(j\omega_{gc})| = 1 \quad (6)$$

and the phase-crossover frequency by

$$\angle L(j\omega_{pc}) = -180^\circ \quad (7)$$

The formal gain and phase margins are

$$GM_{dB} = -20 \log_{10} |L(j\omega_{pc})| \quad (8)$$

$$PM = 180^\circ + \angle L(j\omega_{gc}) \quad (9)$$

The supplied Cadence plots are forward amplifier transfer functions. Therefore, the 0 dB crossing can be reported directly as the unity-gain frequency of the displayed amplifier response, but a formal numerical GM or PM requires a defined feedback-loop transfer function.

3 Part A - Cascode Amplifier with Resistive Load

3.1 Design Calculations

The specified voltage gain is 20 dB. The corresponding linear gain is

$$A_{v,target} = 10^{20/20} = 10 \text{ V/V} \quad (10)$$

A design margin of 20% gives an approximate design gain of

$$A_{v,design} = 1.2(10) = 12 \text{ V/V} \quad (11)$$

For $f_{3dB} = 20 \text{ MHz}$ and $C_L = 5 \text{ pF}$,

$$R_D = \frac{1}{2\pi f_{3dB} C_L} = \frac{1}{2\pi(20 \times 10^6)(5 \times 10^{-12})} = 1591.55 \text{ } \Omega \quad (12)$$

Hence, the selected resistance is

$$R_D \approx 1.6 \text{ k}\Omega \quad (13)$$

From the 1 mW power target,

$$I_D = \frac{P}{V_{DD}} = \frac{1 \times 10^{-3}}{1.8} = 555.6 \text{ } \mu\text{A} \quad (14)$$

For the transistor-width calculations, the handwritten design sheet uses the rounded operating current $I_D = 0.56$ mA.

Using the design gain of approximately 12 V/V,

$$g_m \approx \frac{12}{1.6 \times 10^3} = 7.5 \text{ mS} \quad (15)$$

The corresponding design ratio is

$$\frac{g_m}{I_D} \approx \frac{7.5 \times 10^{-3}}{555.6 \times 10^{-6}} = 13.49 \text{ V}^{-1} \quad (16)$$

3.2 Lower NMOS Characterization and Sizing

The handwritten design sheet specifies approximately $V_{DS} = 0.45$ V and an extracted current density of $I_D/W = 18.60705 \text{ } \mu\text{A}/\mu\text{m}$ for the lower NMOS. Using the rounded sizing current $I_D = 0.56$ mA,

$$W_{N,lower} = \frac{0.56 \text{ mA}}{18.60705 \text{ } \mu\text{A}/\mu\text{m}} = 30.096 \text{ } \mu\text{m} \approx 30 \text{ } \mu\text{m}.$$

Thus, the value used in the final Part-A schematic and in all subsequent Part-A calculations is $W_{N,lower} = 30 \text{ } \mu\text{m}$.

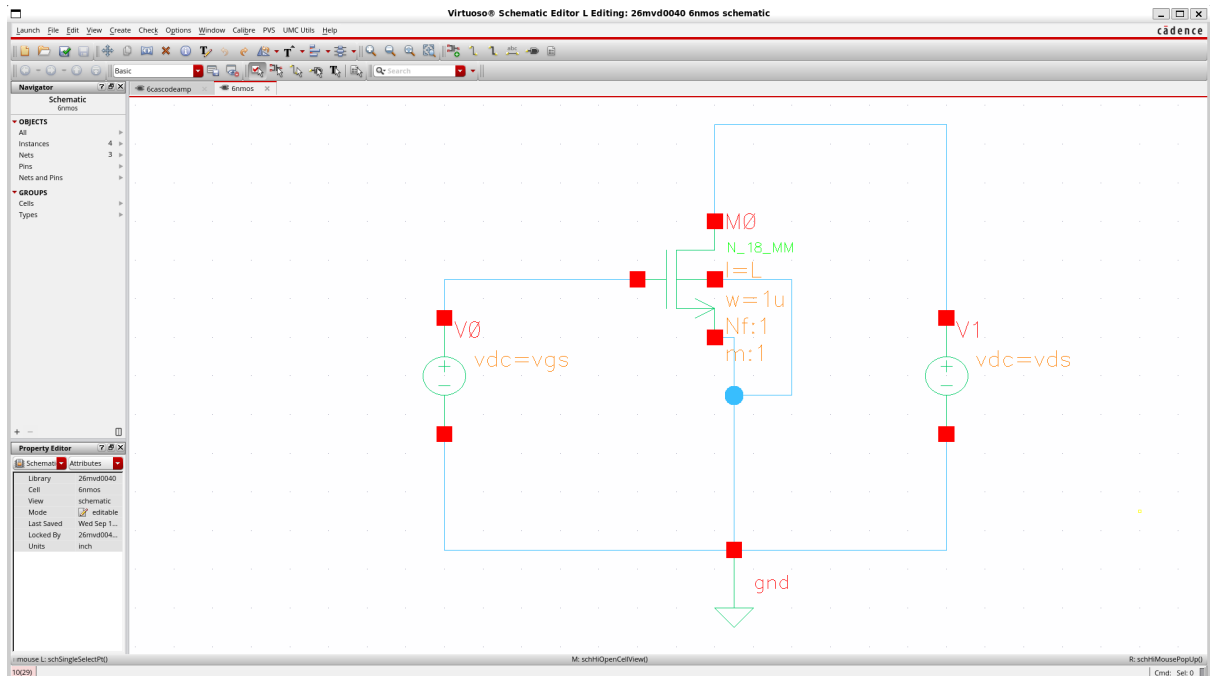


Figure 1: Part A - lower NMOS device-characterization schematic.

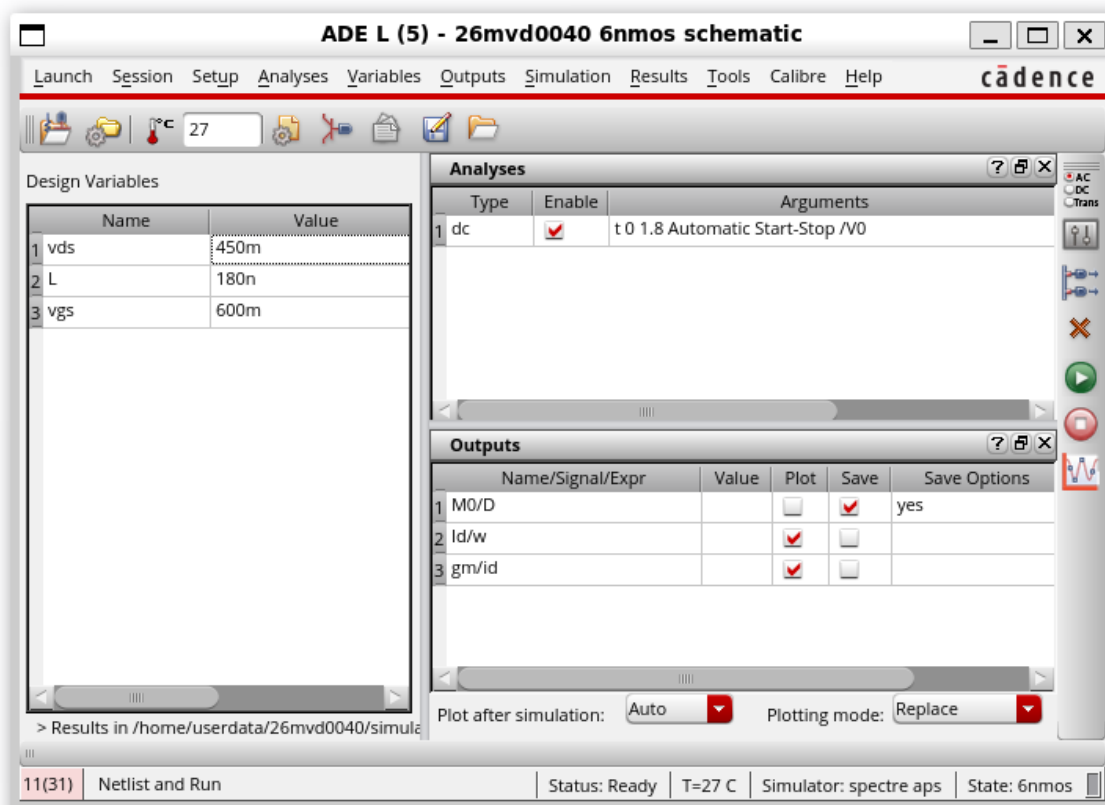


Figure 2: Part A - ADE setup for the lower NMOS characterization.

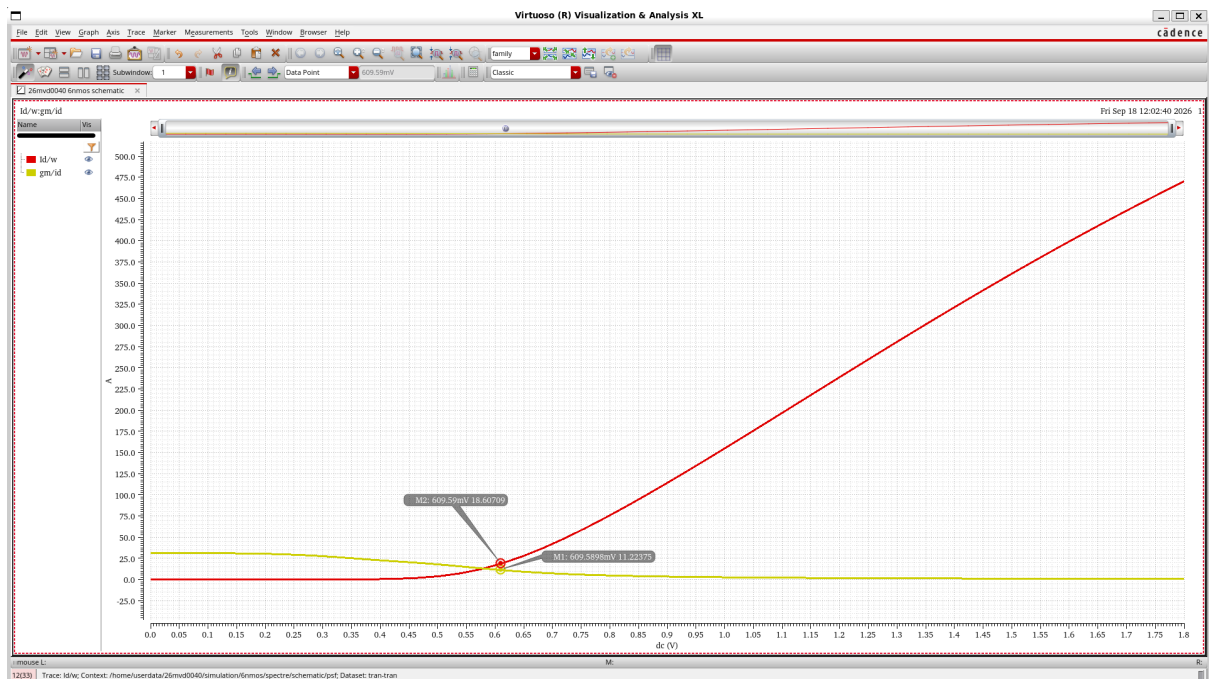


Figure 3: Part A - lower NMOS characterization showing the selected g_m/I_D and I_D/W values.

3.3 Upper NMOS Cascode Device Sizing

The handwritten design sheet gives $g_m/I_D = 11.22375 \text{ V}^{-1}$ and $I_D/W = 18.8739 \mu\text{A}/\mu\text{m}$ for the upper NMOS characterization. Using the same rounded sizing current $I_D = 0.56 \text{ mA}$,

$$W_{N,upper} = \frac{0.56 \text{ mA}}{18.8739 \mu\text{A}/\mu\text{m}} = 29.671 \mu\text{m} \approx 30 \mu\text{m}.$$

Therefore, the implemented upper/cascode NMOS width is also $W_{N,upper} = 30 \mu\text{m}$.

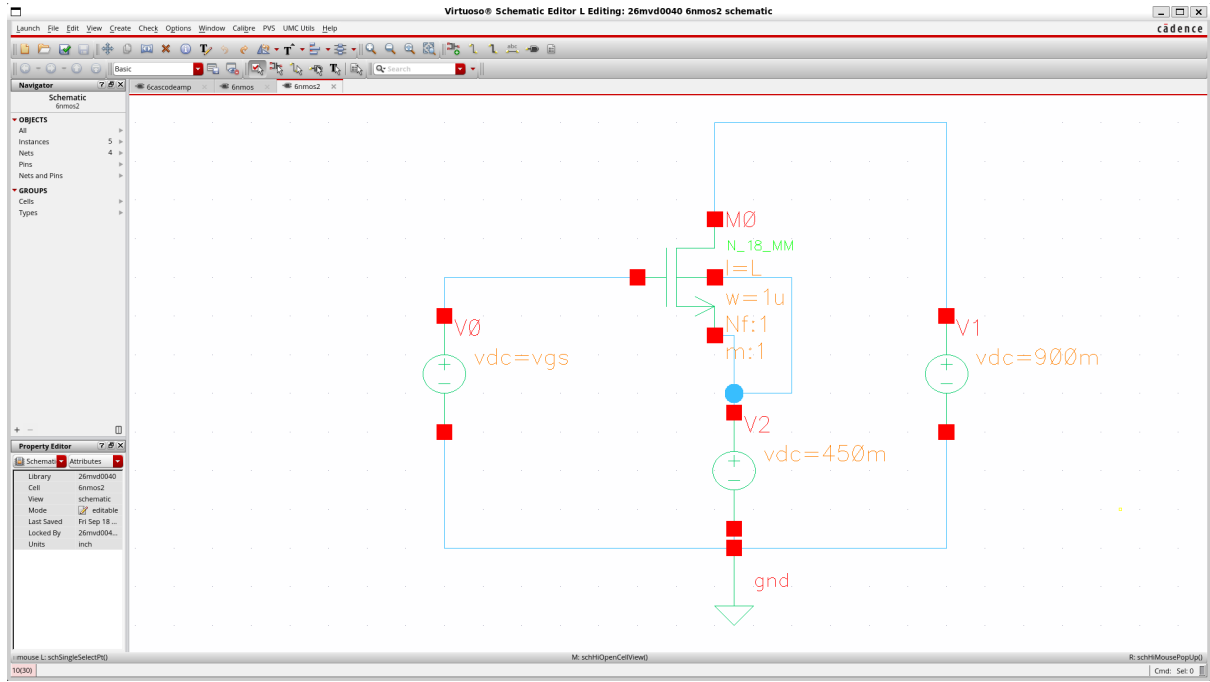


Figure 4: Part A - upper NMOS cascode device-characterization schematic.

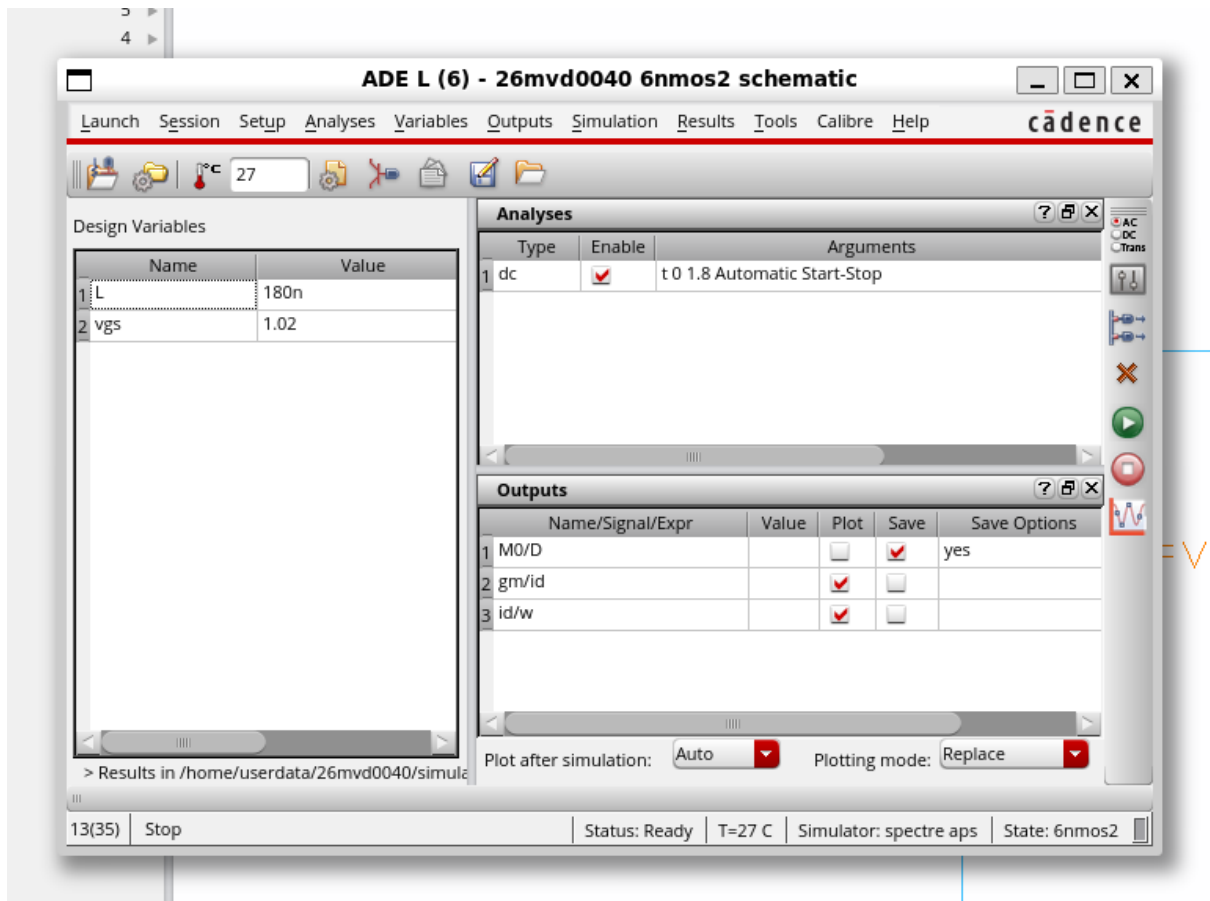


Figure 5: Part A - ADE setup for the upper NMOS characterization.

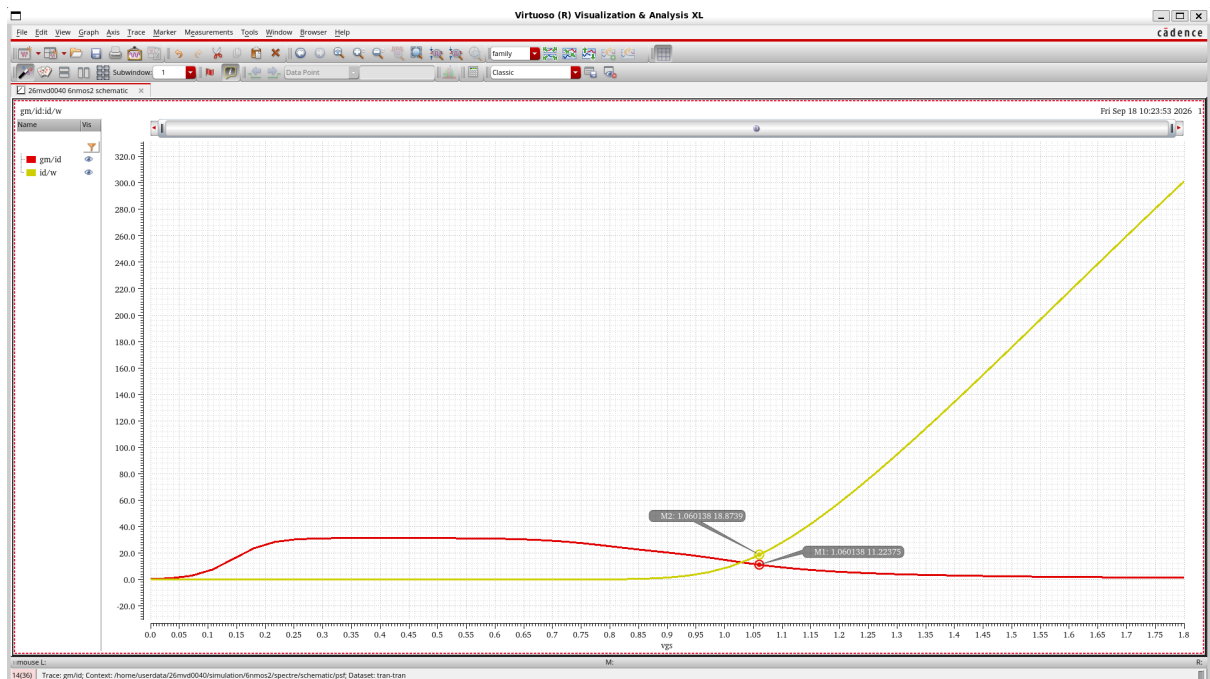


Figure 6: Part A - upper NMOS characterization and selected operating point.

3.4 Final Biasing Conditions

The complete resistive-load circuit uses:

Table 2: Final Part-A circuit parameters

Parameter	Implemented value
V_{DD}	1.8 V
R_D	1.6 k Ω
C_L	5 pF
Lower/input NMOS width	30 μm
Upper/cascode NMOS width	30 μm
NMOS channel length	180 nm
Cascode bias V_B	1.02 V
Input amplitude	10 mV
Input frequency	10 kHz

3.5 Small-Signal Equivalent Circuit

For small-signal analysis, the DC supply and bias sources are treated as AC ground. The cascode stage can therefore be represented using the controlled current sources of the two NMOS devices together with the drain resistance R_D at the output node. The lower-device current source is represented by $g_{m1}v_1$, while the upper cascode device contributes the corresponding controlled-source and body-effect terms. The resulting equivalent circuit used for understanding the cascode gain behaviour is shown below.

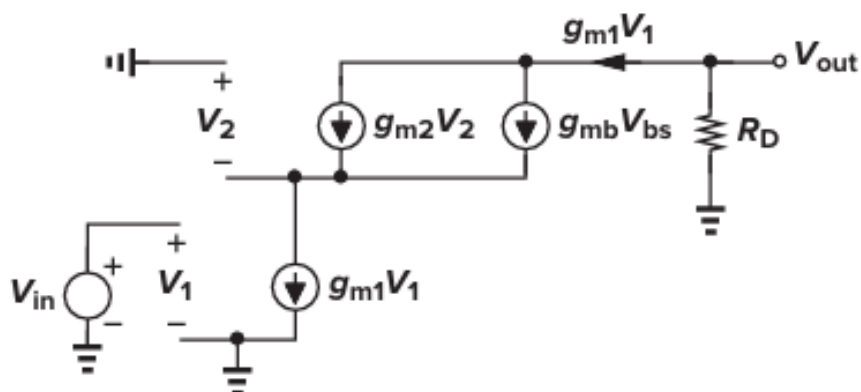


Figure 7: Small-signal equivalent circuit of the cascode stage with resistive load.

The cascode connection increases the effective small-signal output resistance seen at V_{out} . This is the reason the cascoded stage can obtain a larger voltage gain than a single common-source transistor with the same drain resistor.

3.6 Transfer Function of the Small-Signal Cascode Model

For the small-signal circuit shown above, the bias sources are at AC ground. Let v_1 be the internal node between the two NMOS devices. For the ideal small-signal model shown (neglecting r_o and output capacitances), the upper transistor operates as a common-gate device. Its transconductance and body-effect current sources transfer the current produced by the lower transistor to the drain resistor. The node equations may be written as

$$\begin{aligned}(g_{m2} + g_{mb2})v_1 + g_{m1}v_{in} &= 0, \\ v_{out} &= R_D(g_{m2} + g_{mb2})v_1.\end{aligned}$$

Therefore, the transfer function of the displayed ideal small-signal cascode model is

$$H(s) = \frac{v_{out}(s)}{v_{in}(s)} \approx -g_{m1}R_D \quad (17)$$

The result shows that, in the idealized model, the upper cascode device controls the internal node voltage while the low-frequency voltage gain is set by the input-device transconductance and the drain resistance. The body-effect term g_{mb2} affects the internal node voltage but cancels from the ideal voltage-gain expression.

When the load capacitor C_L is included as a first-order output pole, the practical frequency-dependent form becomes

$$H(s) \approx \frac{-g_{m1}R_D}{1 + sR_DC_L} \quad (18)$$

with the corresponding dominant pole

$$\omega_p \approx \frac{1}{R_DC_L}, \quad f_p \approx \frac{1}{2\pi R_DC_L}. \quad (19)$$

For the implemented values $R_D = 1.6 \text{ k}\Omega$ and $C_L = 5 \text{ pF}$, this first-order estimate gives $f_p \approx 19.89 \text{ MHz}$, which is close to the designed 20 MHz bandwidth target. The Cadence pole-zero and AC plots presented later in this report provide the simulated values for the complete transistor-level circuit.

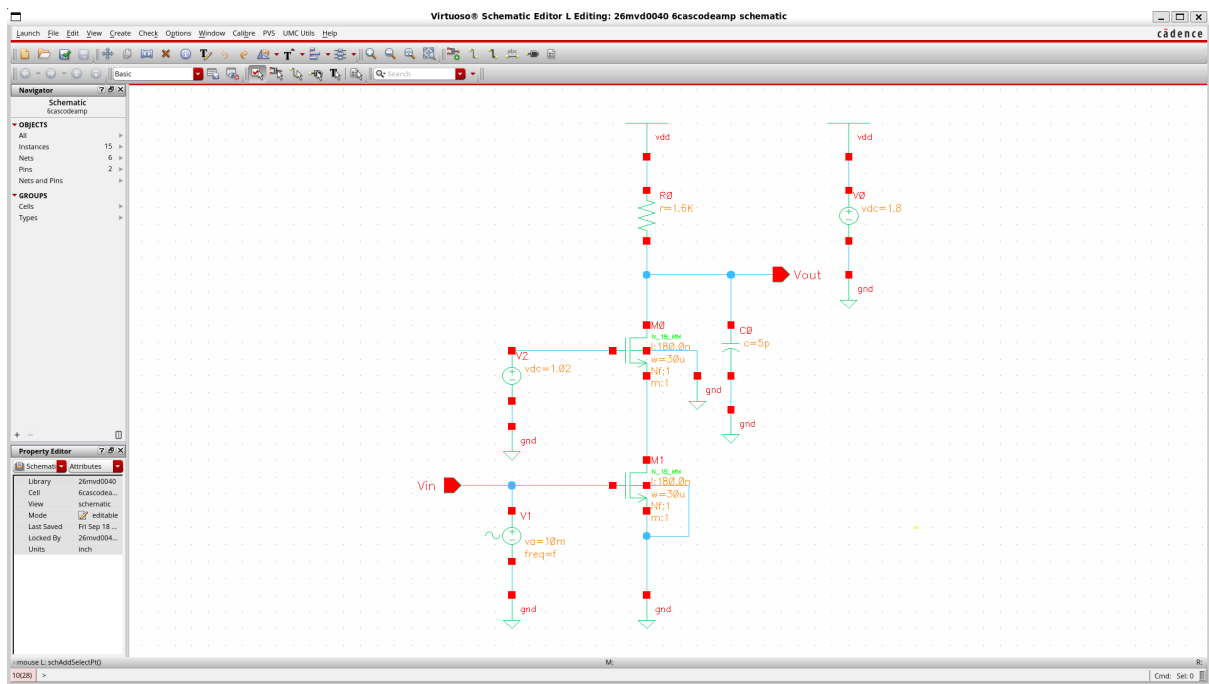


Figure 8: Part A - complete resistive-load cascode schematic.

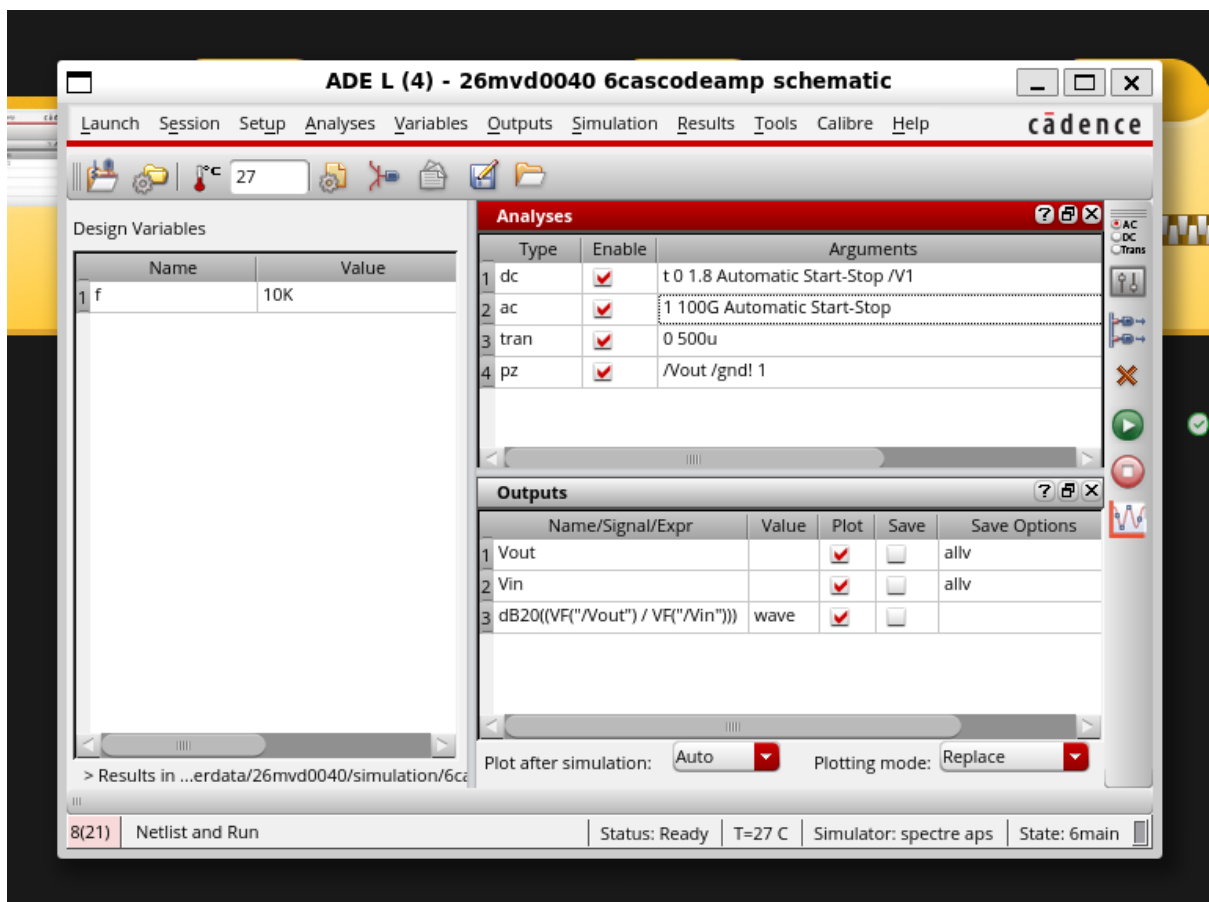


Figure 9: Part A - ADE final analysis setup.

3.7 DC and Transient Analysis

The combined Cadence output shows the DC transfer characteristic, transient response and the corresponding AC response. The transient waveform is inverted as expected from the common-source input path. The output waveform swings over a considerably larger range than the input signal, with the exact limits set by the nonlinear operating region of the implemented circuit.

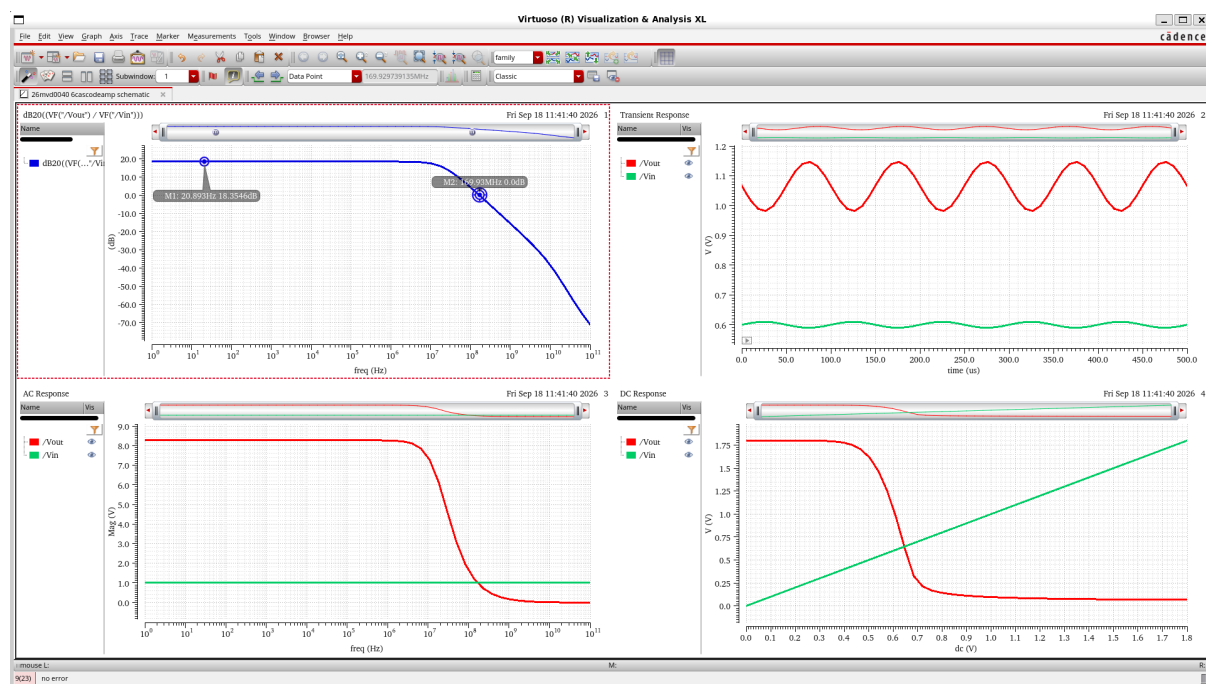


Figure 10: Part A - combined DC, transient and AC simulation output.

3.8 Pole-Zero Analysis

The submitted PZ result gives two real poles and one real zero. The extracted values are

$$\begin{aligned} p_1 &= -2.0087 \times 10^7 \text{ Hz}, \\ p_2 &= -1.02605 \times 10^{10} \text{ Hz}, \\ z_1 &= +7.43895 \times 10^{10} \text{ Hz}. \end{aligned}$$

The reported DC gain is 8.27430 V/V. Its corresponding gain in decibels is approximately

$$A_{DC,dB} = 20 \log_{10}(8.27430) \approx 18.35 \text{ dB} \quad (19)$$

The measured first pole therefore indicates a 3-dB bandwidth of approximately 20.09 MHz for the dominant-pole portion of the response.

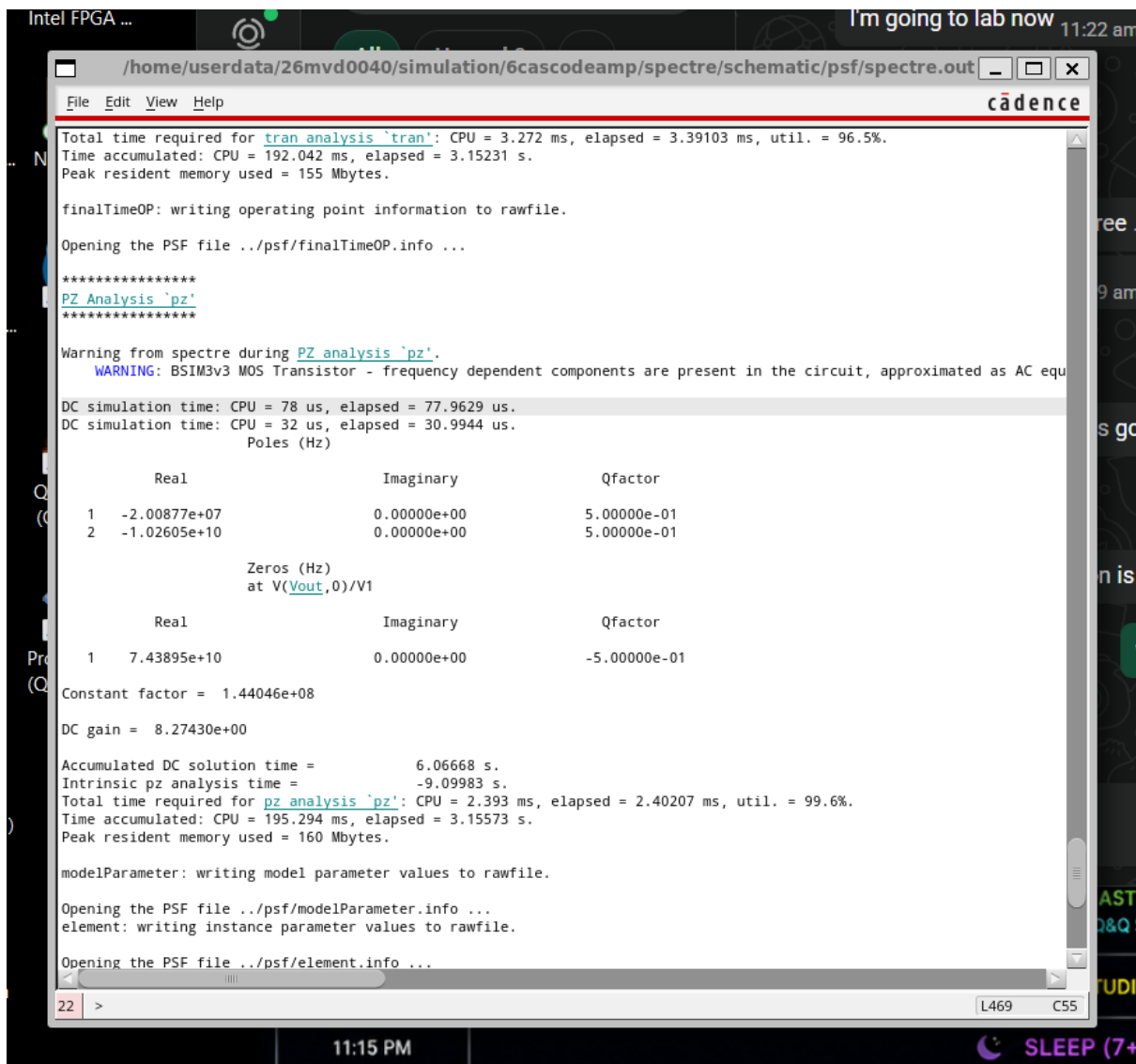


Figure 11: Part A - submitted Cadence pole-zero analysis output.

3.9 AC Gain Response

The final dB plot marks a low-frequency gain of 18.3546 dB at 724.436 Hz and the 0 dB crossing at 169.93 MHz. The unity-gain frequency is therefore approximately 169.93 MHz.

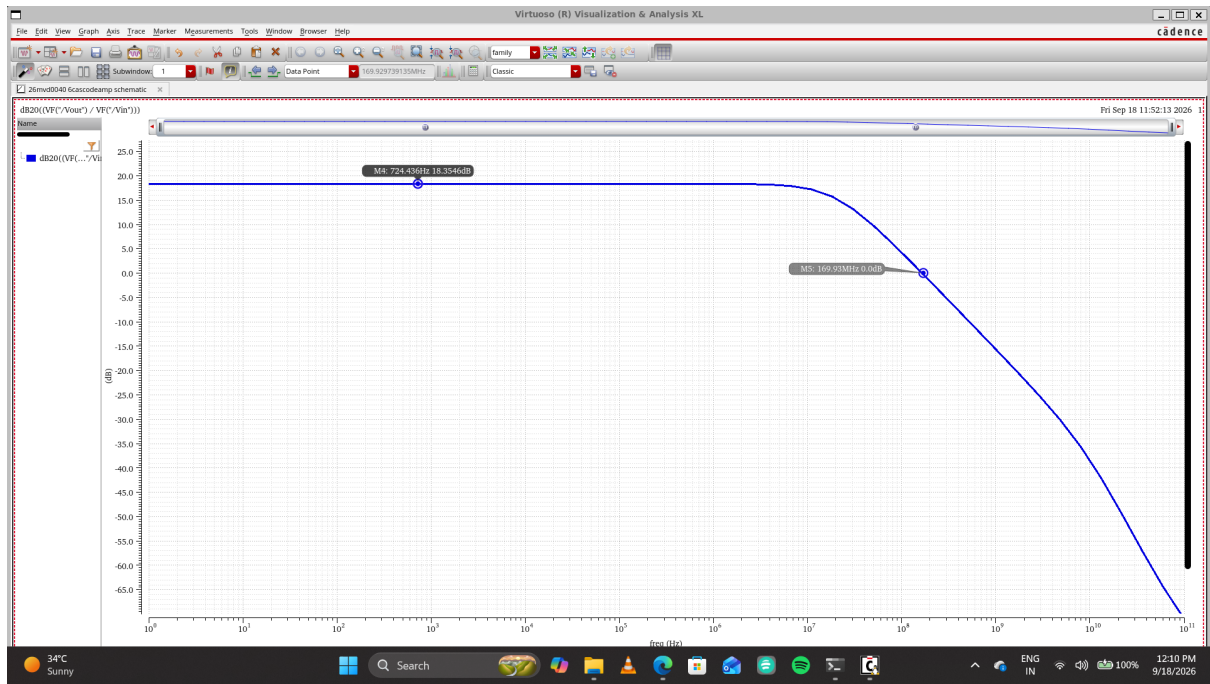


Figure 12: Part A - final gain response with low-frequency and 0 dB markers.

Table 3: Part-A final simulated values

Quantity	Measured / extracted value
PZ DC gain	8.27430 V/V
Low-frequency gain	18.3546 dB
Target gain	20 dB
Gain difference	1.6454 dB
Dominant pole	-2.0087×10^7 Hz
Second pole	-1.02605×10^{10} Hz
Zero	$+7.43895 \times 10^{10}$ Hz
Approx. 3-dB bandwidth from dominant pole	20.09 MHz
Unity-gain frequency	169.93 MHz

Relative to the 20 dB target, the final low-frequency gain is lower by 1.6454 dB. In linear voltage-gain terms, the extracted gain is 8.27430 V/V rather than the ideal 10 V/V target.

4 Part B - Cascode Amplifier with PMOS Cascode Current-Source Load

4.1 Objective and Biasing Approach

Part B replaces the resistive load with a PMOS cascode current-source load. Four devices are characterized separately: the lower NMOS, upper NMOS, lower PMOS and upper PMOS. The submitted width sweeps are used directly to select a width that produces approximately $100\text{ }\mu\text{A}$ of drain-current magnitude.

The handwritten task sheet specifies a PMOS current-source-loaded cascode with $I_{bias} = 100\text{ }\mu\text{A}$ and a voltage-gain requirement greater than 20 dB. The submitted PDM/width-sweep results are therefore used directly for the four transistor widths. The final Part-B circuit uses $V_{DD} = 1.8\text{ V}$, $C_L = 5\text{ pF}$, a 10 mV input at 10 kHz, an NMOS bias near 1.02 V, a lower-PMOS gate bias of 0.75 V and an upper-PMOS gate bias of 1.2 V. The transistor channel length shown in the submitted schematics is 180 nm.

4.2 Lower NMOS Width Sweep

The lower NMOS width sweep gives the marker

$$W_{N,lower} = 6.469202\text{ }\mu\text{m} \quad \text{at} \quad I_D = +100\text{ }\mu\text{A} \quad (20)$$

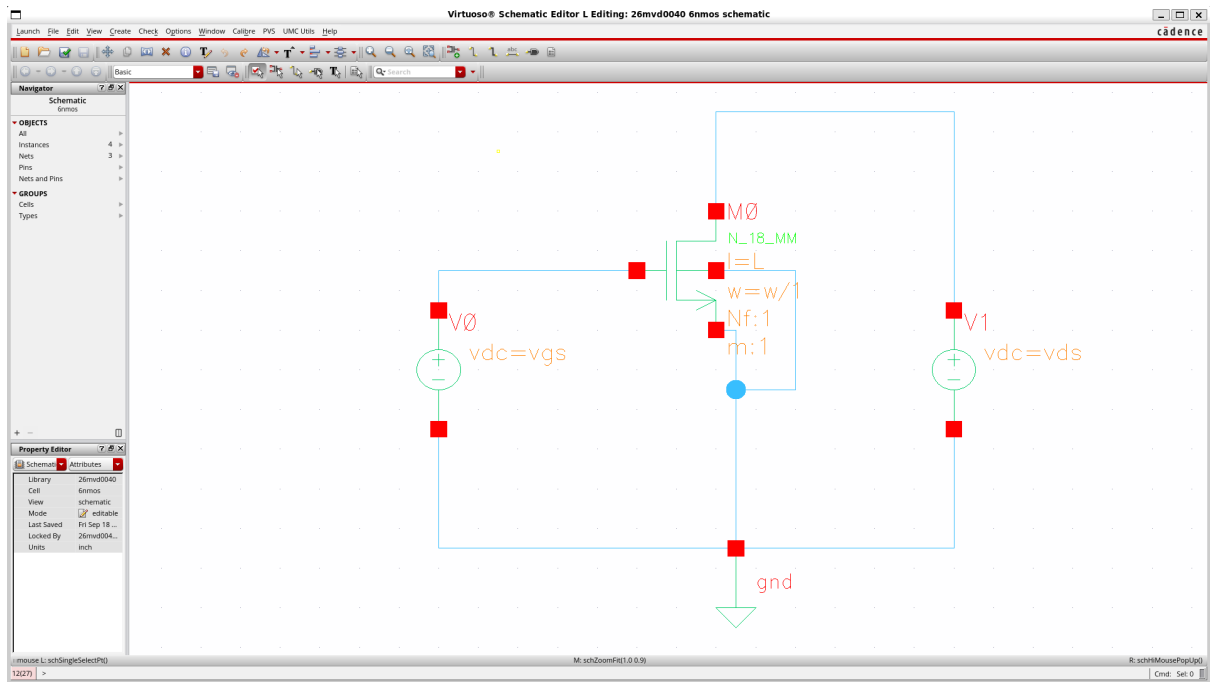


Figure 13: Part B - lower NMOS width-sweep schematic.

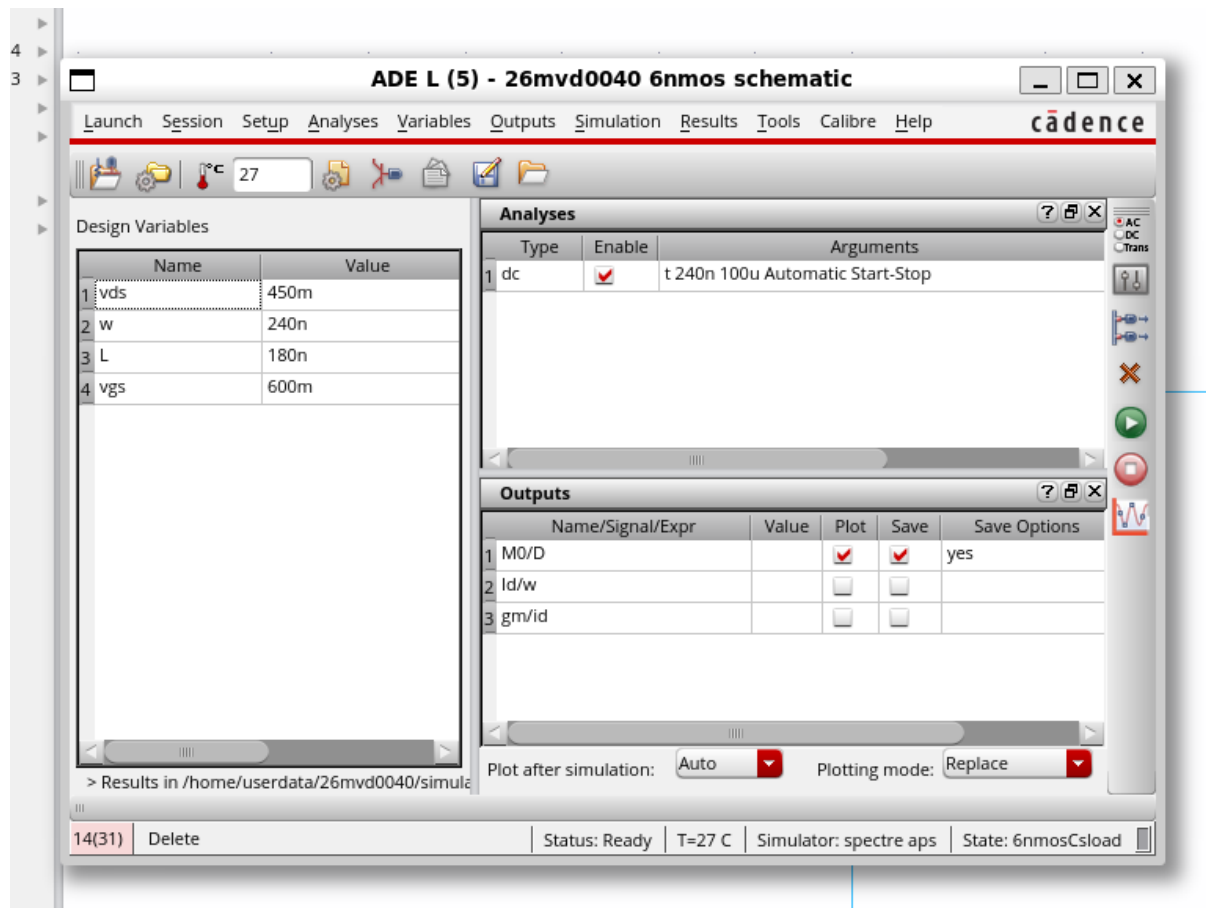


Figure 14: Part B - lower NMOS ADE setup.

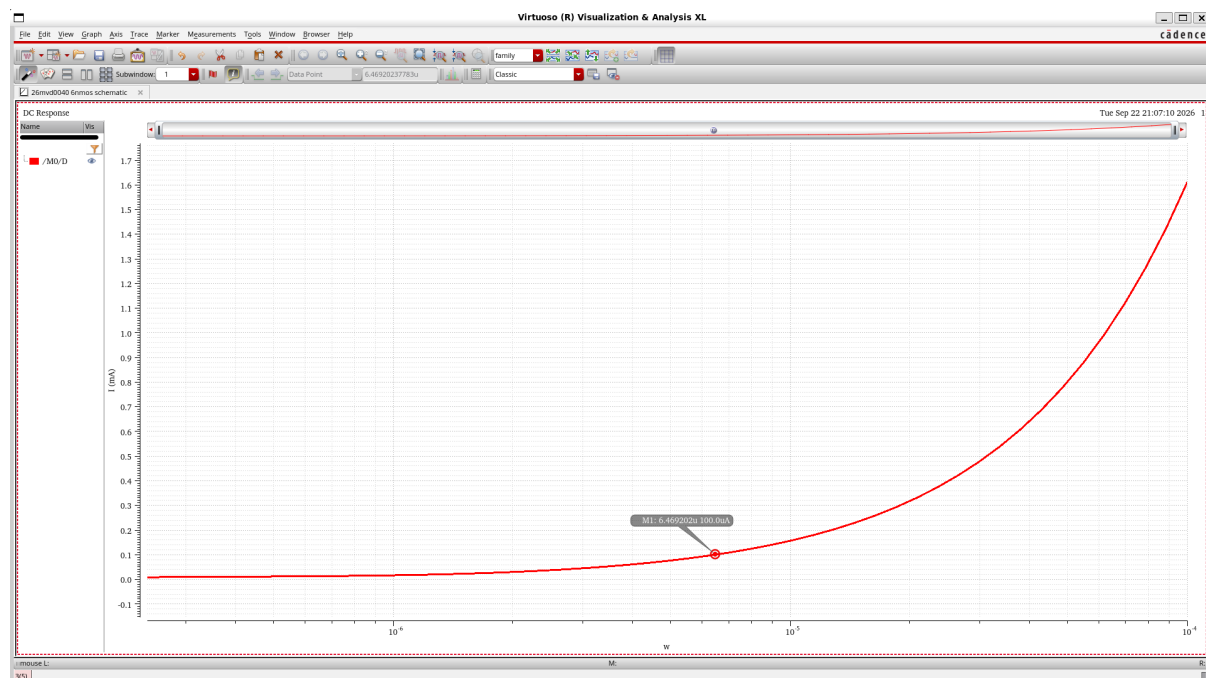


Figure 15: Part B - lower NMOS drain-current versus width response.

Two additional Cadence captures from the same lower-NMOS characterization sequence are retained in the project and are shown again in the appendix.

4.3 Upper NMOS Width Sweep

The upper NMOS sweep gives the marker

$$W_{N,upper} = 9.433895 \mu\text{m} \quad \text{at} \quad I_D = +100 \mu\text{A} \quad (21)$$

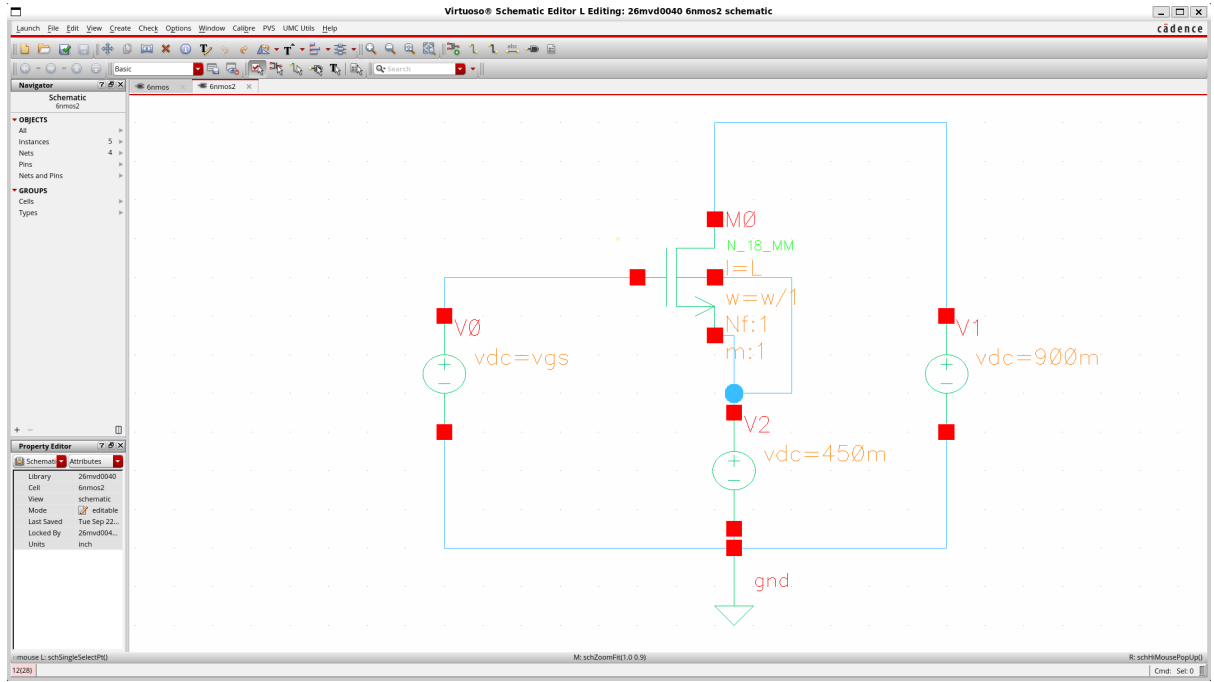


Figure 16: Part B - upper NMOS width-sweep schematic.

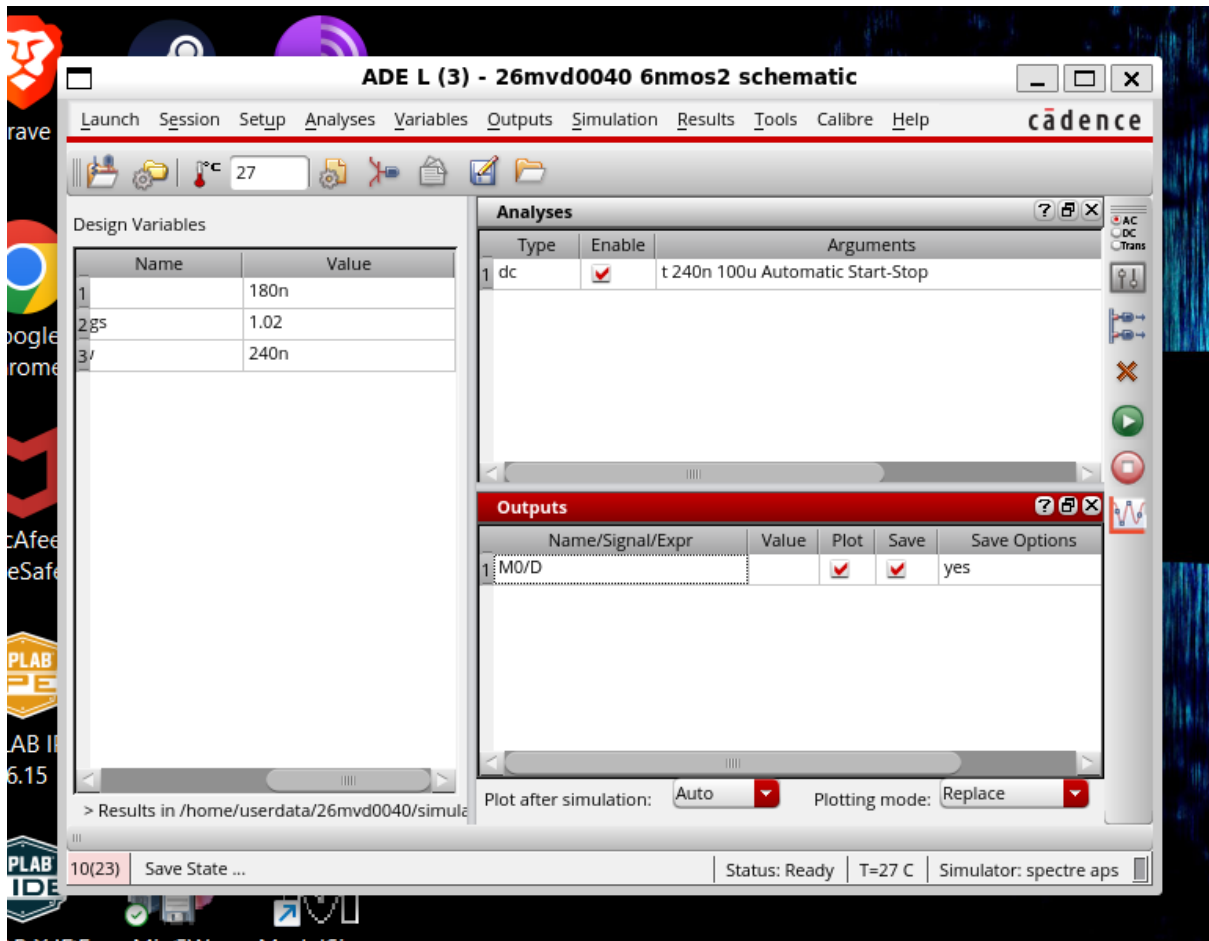


Figure 17: Part B - upper NMOS ADE setup.

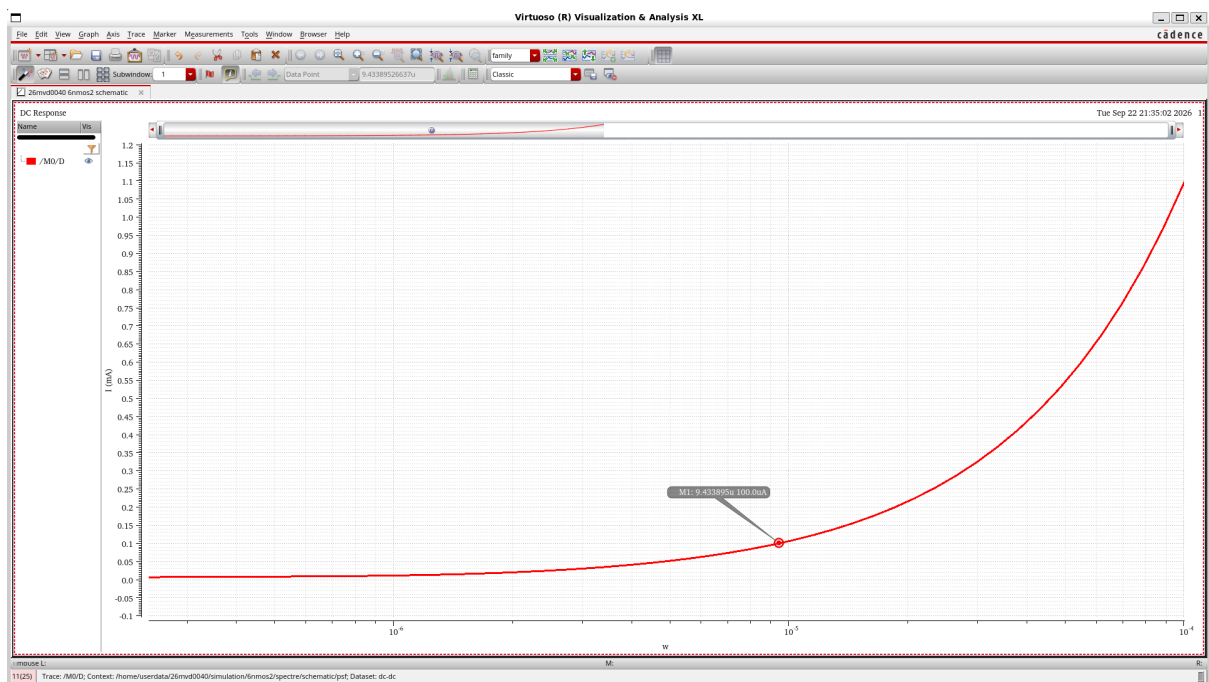


Figure 18: Part B - upper NMOS drain-current versus width response.

The upper-NMOS test uses approximately 450 mV of drain-to-source voltage, as shown by the supplied characterization schematic.

4.4 Lower PMOS Width Sweep

The lower PMOS sweep gives

$$W_{P,lower} = 20.415090 \mu\text{m} \quad \text{at} \quad I_D = -100 \mu\text{A} \quad (22)$$

The negative sign is due to the PMOS current-direction convention used in the Cadence plot.

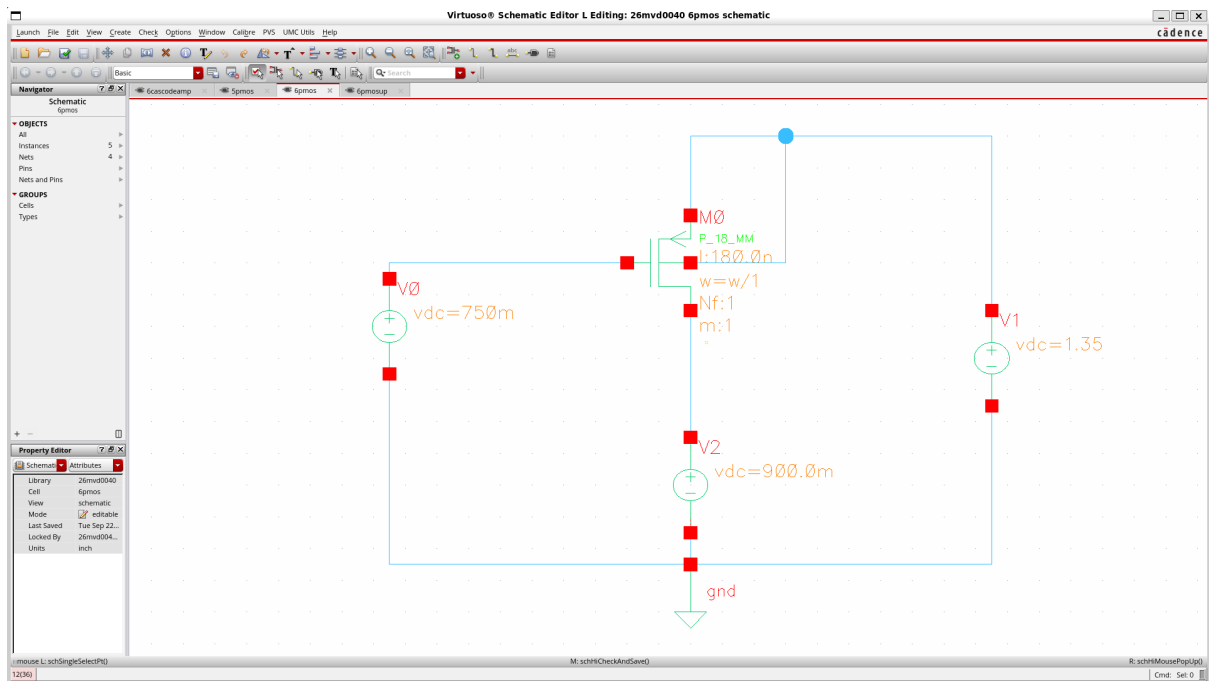


Figure 19: Part B - lower PMOS width-sweep schematic.

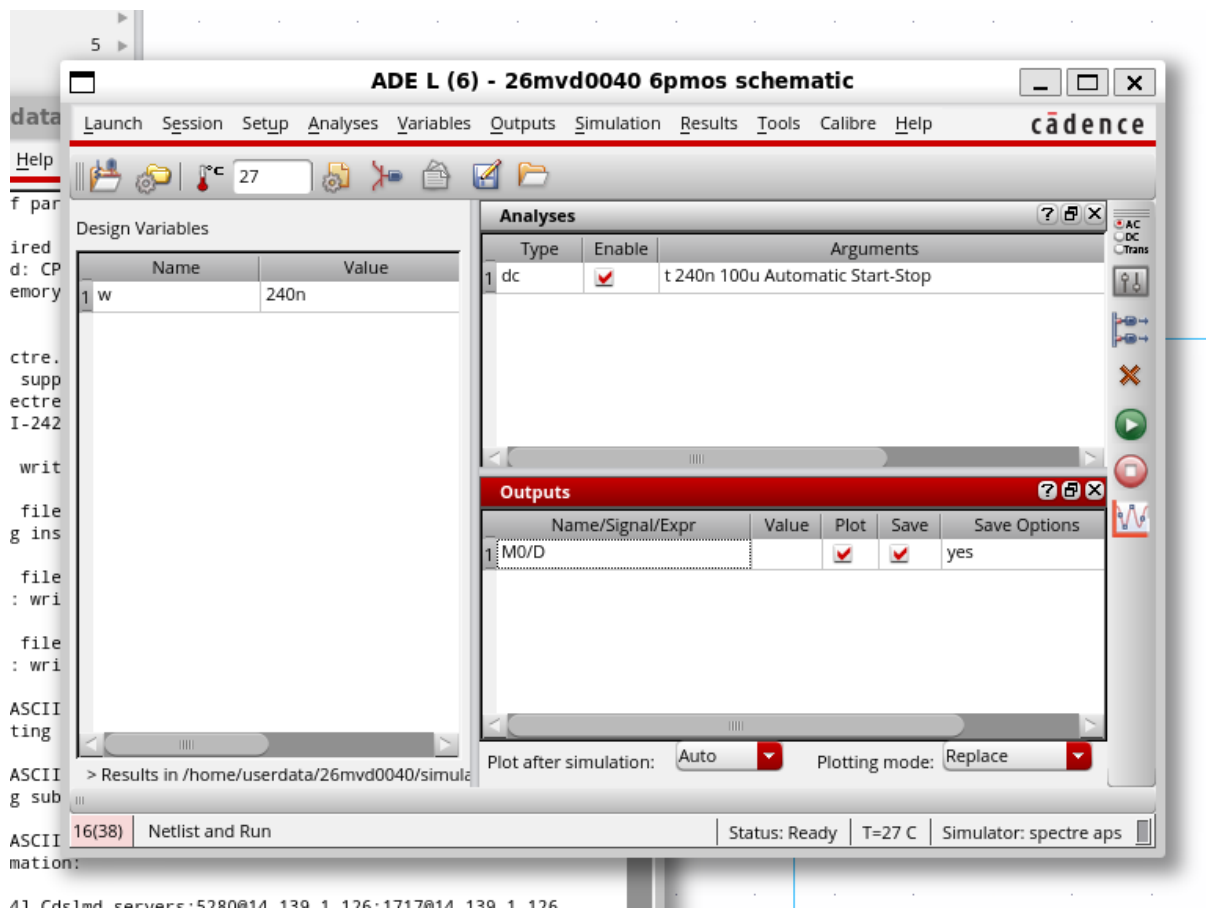


Figure 20: Part B - lower PMOS ADE setup.

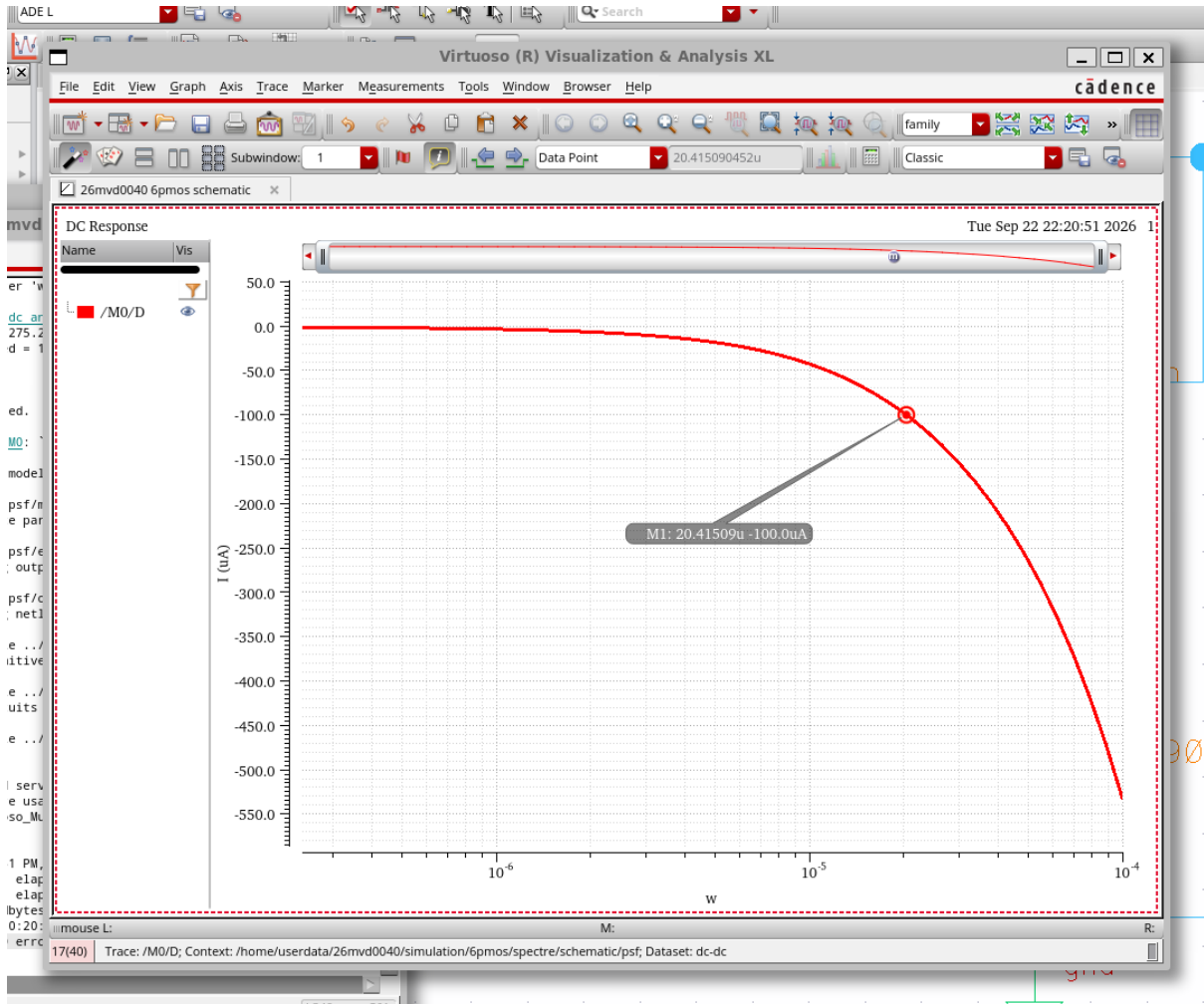


Figure 21: Part B - lower PMOS drain-current versus width response.

4.5 Upper PMOS Width Sweep

The upper PMOS sweep gives the same selected width

$$W_{P,upper} = 20.415090 \mu\text{m} \quad \text{at} \quad I_D = -100 \mu\text{A} \quad (23)$$

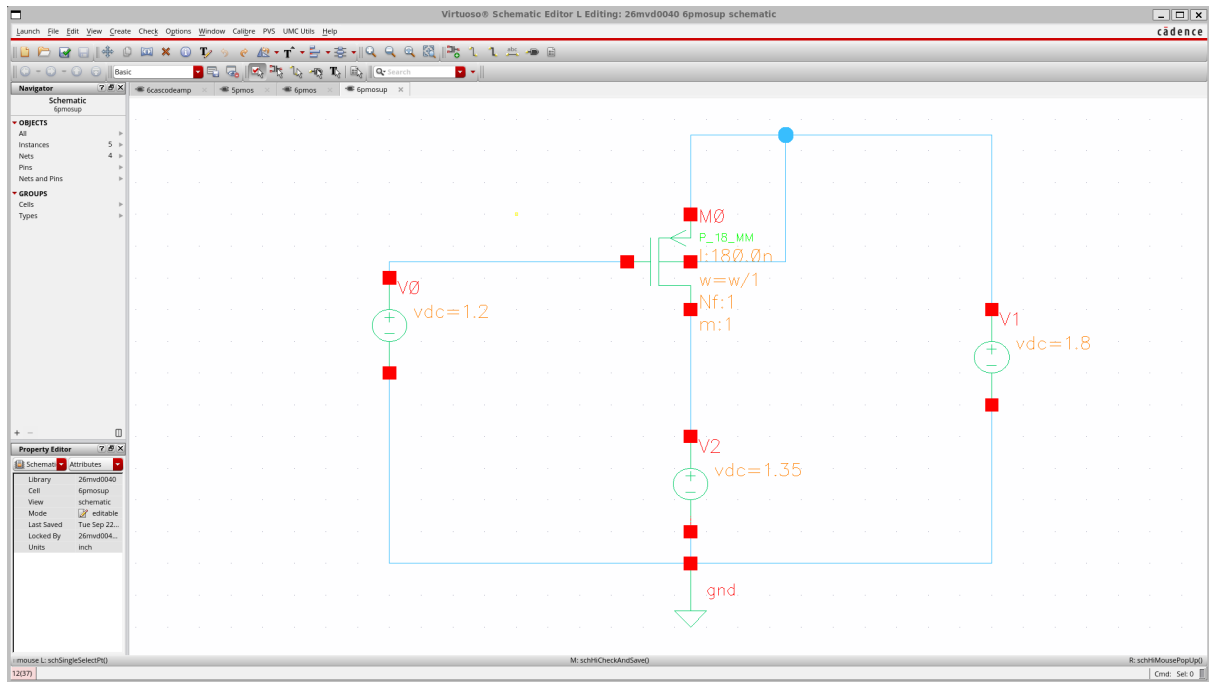


Figure 22: Part B - upper PMOS width-sweep schematic.

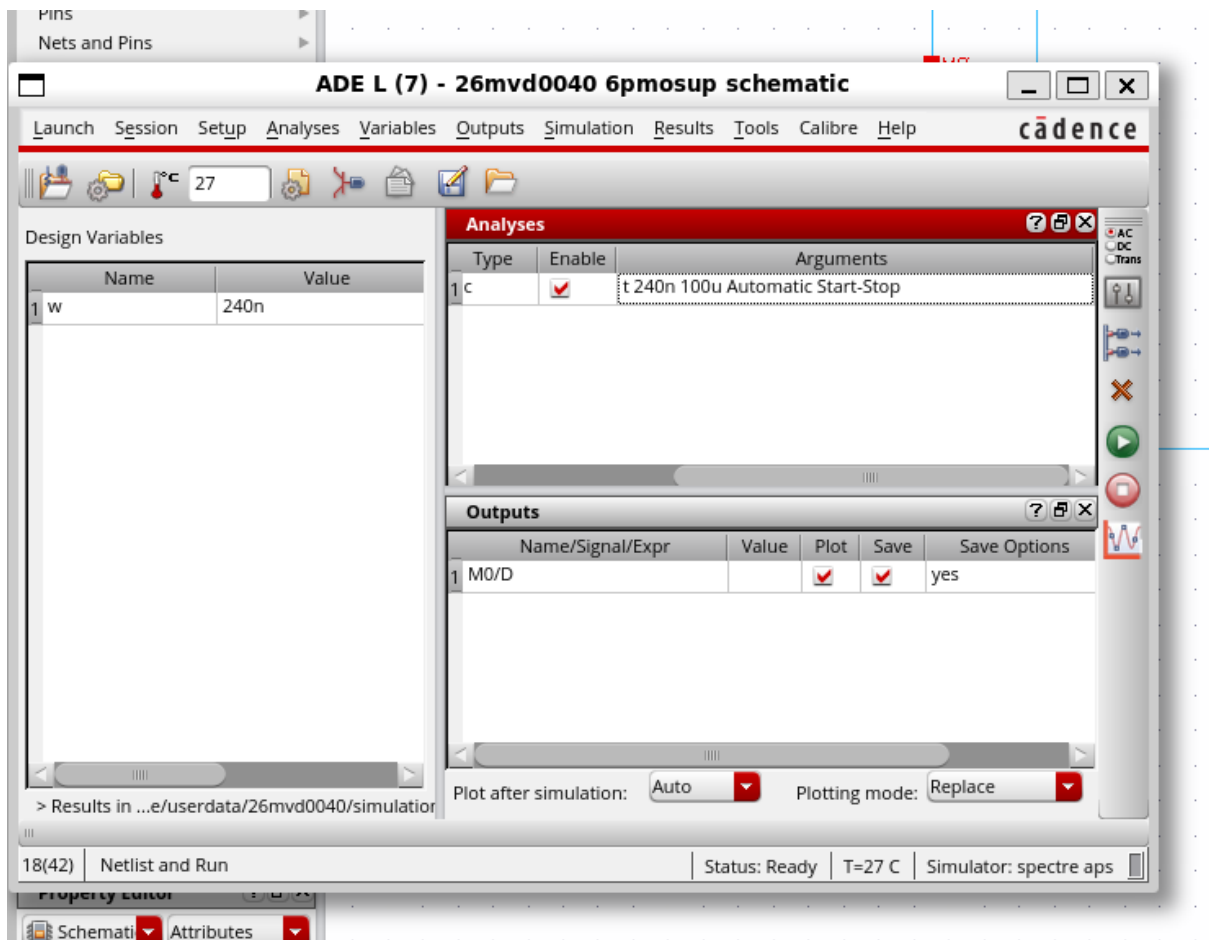


Figure 23: Part B - upper PMOS ADE setup.

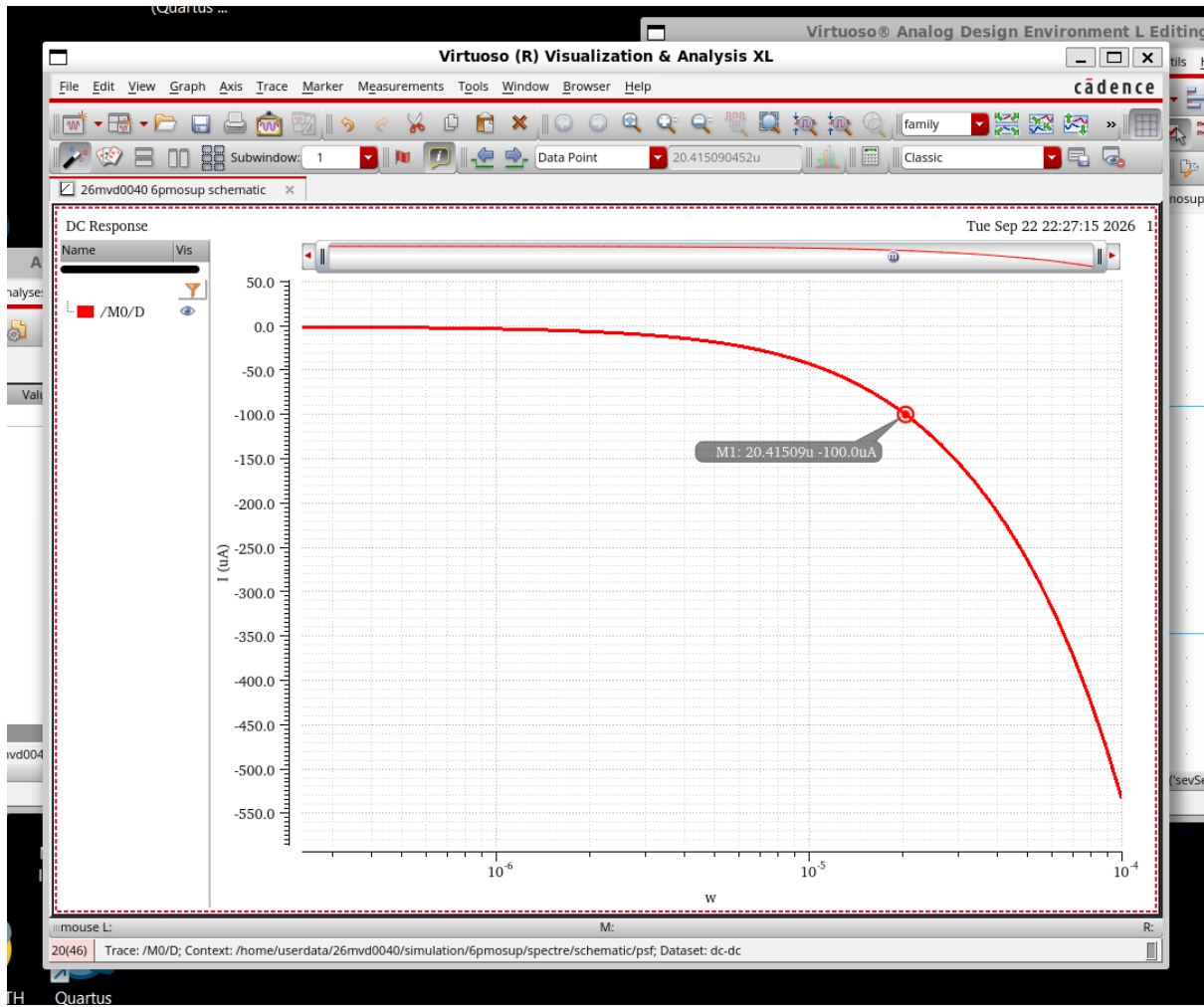


Figure 24: Part B - upper PMOS drain-current versus width response.

4.6 Part-B Transistor Width Summary

Table 4: Transistor widths selected from the submitted Part-B width sweeps

Transistor	Sweep condition	Selected width
Lower NMOS	+100 μA	6.469202 μm
Upper NMOS	+100 μA	9.433895 μm
Lower PMOS	-100 μA	20.415090 μm
Upper PMOS	-100 μA	20.415090 μm

4.7 Complete PMOS Cascode Current-Source Load Circuit

The final schematic consists of two NMOS transistors in a lower cascode amplifying stack and two PMOS transistors in the upper cascode current-source load. The output is taken from the common node between the two stacks. The submitted schematic labels the transistor widths as approximately $6.47\text{ }\mu\text{m}$ and $9.43\text{ }\mu\text{m}$ for the NMOS devices and $20.42\text{ }\mu\text{m}$ for each PMOS device.

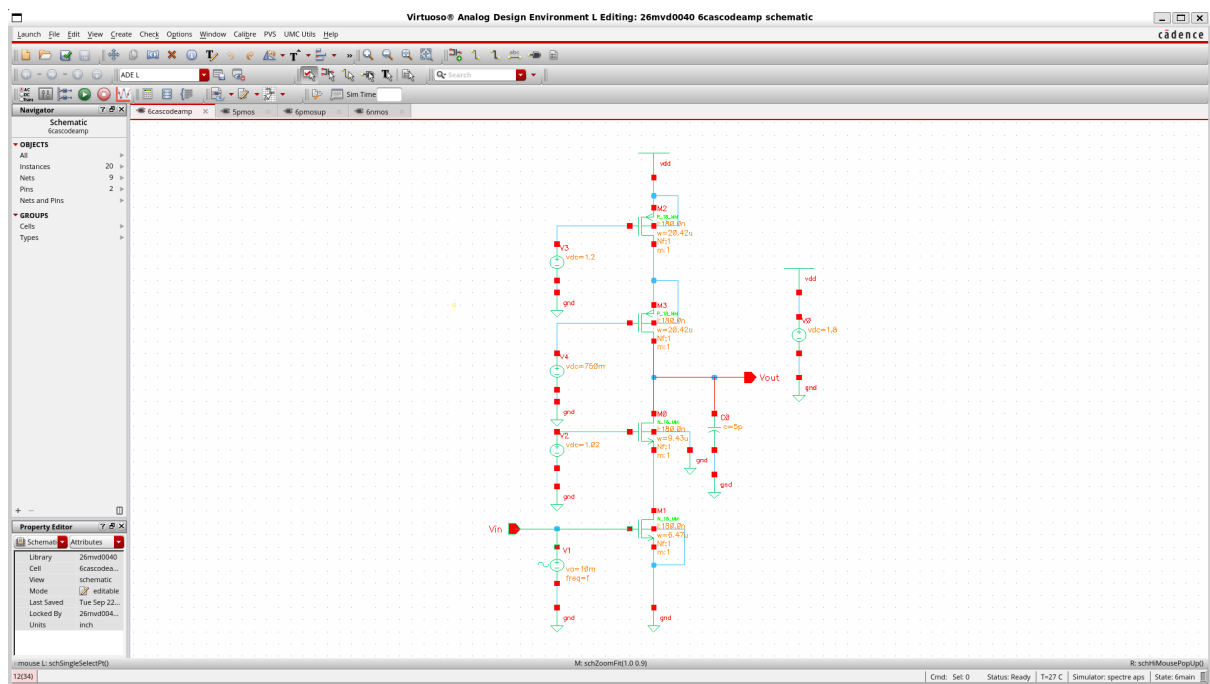


Figure 25: Part B - complete PMOS cascode current-source loaded amplifier.

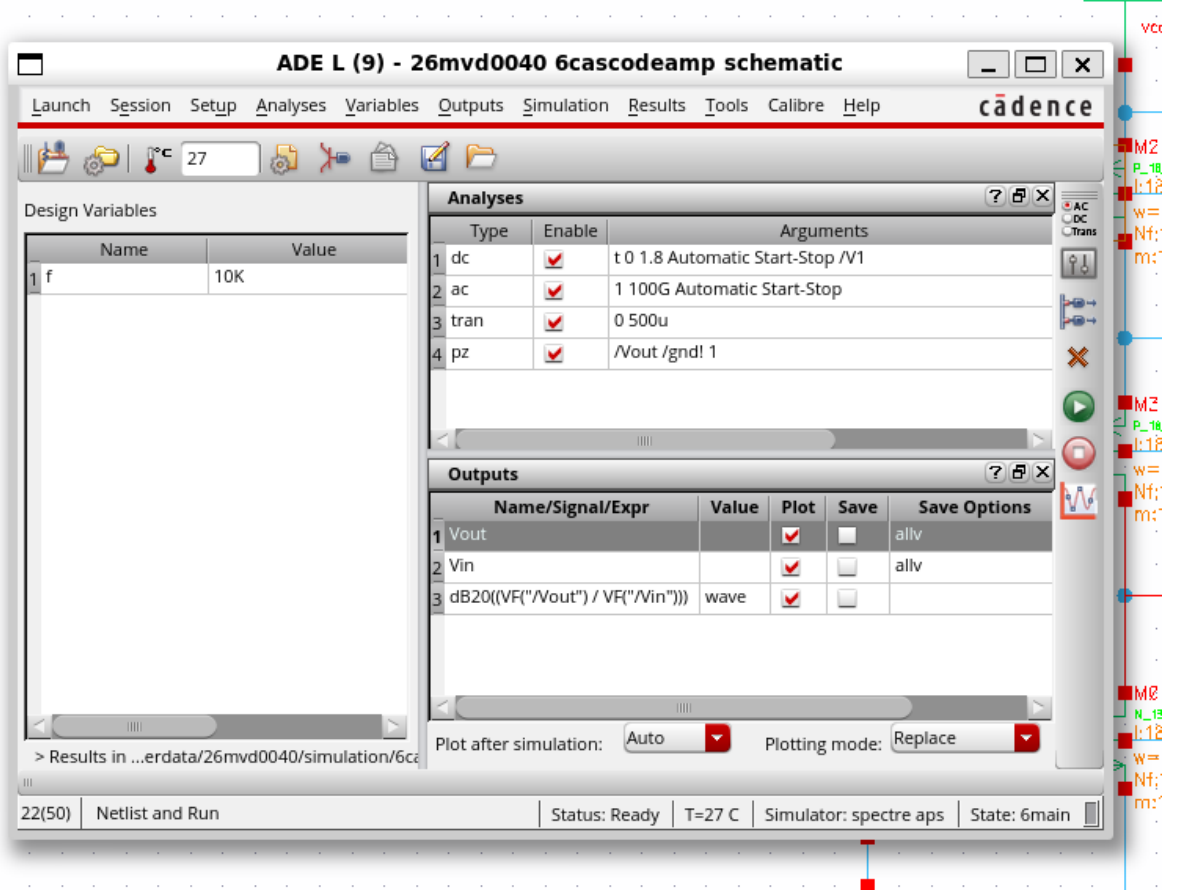


Figure 26: Part B - ADE final analysis setup.

4.8 Small-Signal Model and Transfer Function

For Part B, the two NMOS transistors form the lower cascode amplifying stage and the two PMOS transistors form the cascode current-source load. In the small-signal model, the effective output resistance is the parallel combination of the NMOS and PMOS cascode output resistances. Let

$$R_{out} = R_{out,N} \parallel R_{out,P}$$

and let the total capacitance at the output node be $C_{out} = C_L + C_{par}$, where C_{par} represents the transistor parasitic capacitances. A compact first-order transfer function is therefore

$$H_B(s) = \frac{V_{out}(s)}{V_{in}(s)} \approx \frac{-g_{m,n}R_{out}}{1 + sR_{out}C_{out}} \quad (24)$$

At frequencies well below the first pole, the low-frequency gain is

$$A_{v0} \approx -g_{m,n}R_{out} \quad (25)$$

The high output resistance produced by the NMOS and PMOS cascode structures is responsible for the large voltage gain of the active-load implementation.

For the submitted Cadence pole-zero result, the extracted poles are

$$\begin{aligned} p_1 &= -7.88371 \times 10^5 \text{ Hz}, \\ p_2 &= -3.14795 \times 10^9 \text{ Hz}, \\ p_3 &= -9.24943 \times 10^9 \text{ Hz}, \end{aligned}$$

and the extracted zeros are

$$\begin{aligned} z_1 &= -3.14453 \times 10^9 \text{ Hz}, \\ z_2 &= +7.42947 \times 10^{10} \text{ Hz}. \end{aligned}$$

Using the measured DC gain magnitude of 45.6276 V/V, the corresponding normalized measured transfer function can be written as

$$H_B(s) \approx -45.6276 \frac{\left(1 + \frac{s}{2\pi(3.14453 \times 10^9)}\right) \left(1 - \frac{s}{2\pi(7.42947 \times 10^{10})}\right)}{\left(1 + \frac{s}{2\pi(7.88371 \times 10^5)}\right) \left(1 + \frac{s}{2\pi(3.14795 \times 10^9)}\right) \left(1 + \frac{s}{2\pi(9.24943 \times 10^9)}\right)}$$

The first pole is the dominant pole of the measured response, while the higher-frequency poles and zeros describe the additional high-frequency behaviour of the implemented PMOS current-source-loaded cascode.

4.9 DC, Transient and AC Analysis

The supplied combined result shows the DC transfer curve, transient response and AC magnitude response. The DC characteristic exhibits the expected transition between the low-output and high-output regions around the bias point. The transient waveform is strongly amplified and limited by the circuit operating range. The AC response shows a substantially higher midband gain than the resistive-load implementation.

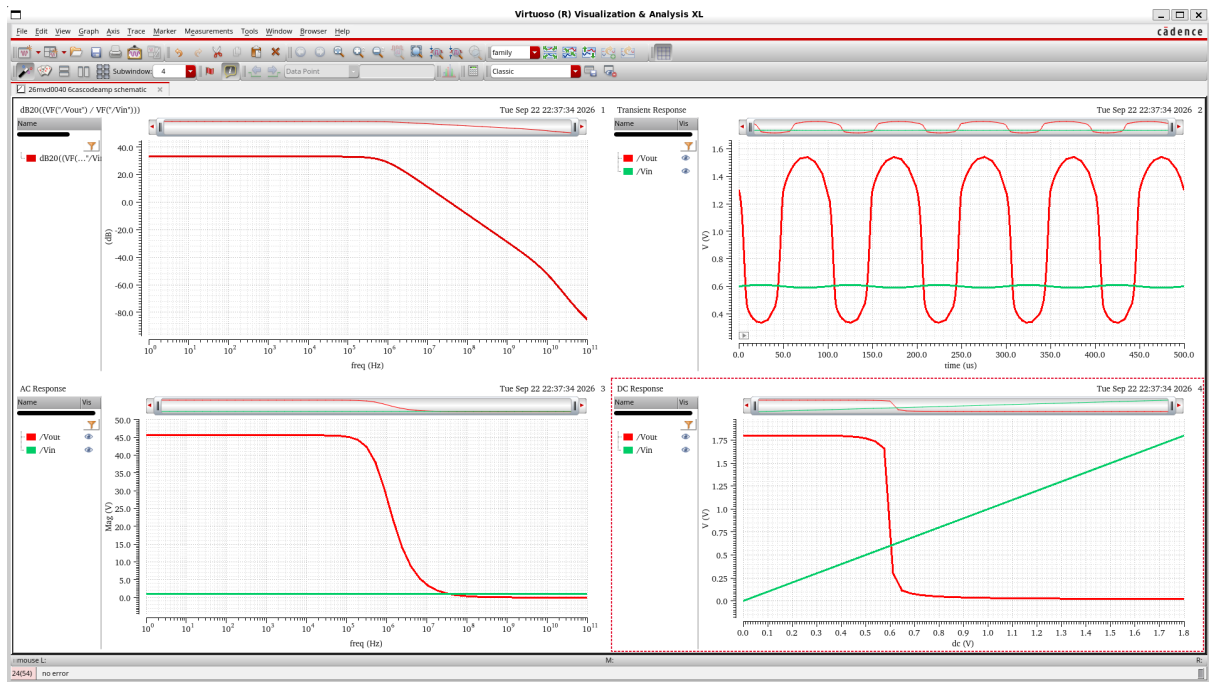


Figure 27: Part B - combined DC, transient and AC simulation output.

4.10 Final AC Gain Response

The final Part-B dB response gives a low-frequency gain marker of 33.1832 dB at 724.436 Hz. The 0 dB crossing is at 37.0662 MHz.

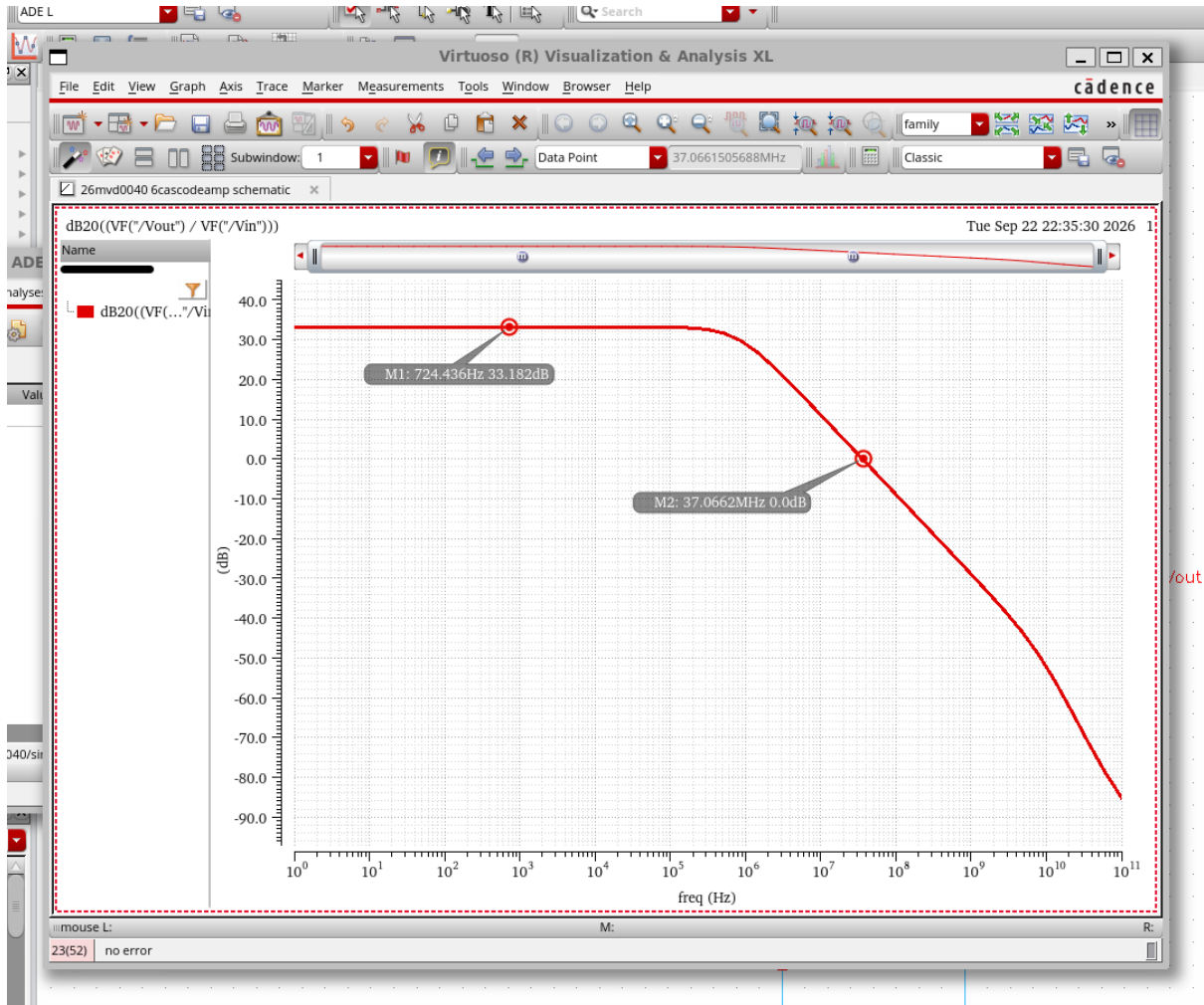


Figure 28: Part B - final gain response with low-frequency and 0 dB markers.

The corresponding PZ analysis gives a DC gain of 45.6276 V/V. The measured low-frequency gain and the PZ result are consistent at approximately 33.18 dB.

4.11 Pole-Zero Analysis

The additional submitted PZ output, which was not included in the main archive, is included here as requested. The extracted values are

$$p_1 = -7.88371 \times 10^5 \text{ Hz},$$

$$p_2 = -3.14795 \times 10^9 \text{ Hz},$$

$$p_3 = -9.24943 \times 10^9 \text{ Hz},$$

$$z_1 = -3.14453 \times 10^9 \text{ Hz},$$

$$z_2 = +7.42947 \times 10^{10} \text{ Hz}.$$

The reported constant factor is 2.81687×10^7 and the DC gain is 45.6276 V/V. The first pole is the dominant pole and corresponds to an approximately 788.371 kHz first-order

3-dB bandwidth estimate.

Poles (Hz)			
	Real	Imaginary	Qfactor
1	-7.88371e+05	0.00000e+00	5.00000e-01
2	-3.14795e+09	0.00000e+00	5.00000e-01
3	-9.24943e+09	0.00000e+00	5.00000e-01
Zeros (Hz) at V(Vout ,0)/V1			
	Real	Imaginary	Qfactor
1	-3.14453e+09	0.00000e+00	5.00000e-01
2	7.42947e+10	0.00000e+00	-5.00000e-01
Constant factor = 2.81687e+07			
DC gain = 4.56276e+01			

Figure 29: Part B - additional pole-zero analysis output supplied separately from the main archive.

4.12 Measured Frequency-Response Values

Table 5: Part-B final frequency-response values

Parameter	Measured / extracted value
PZ DC gain	45.6276 V/V
Low-frequency gain	33.1832 dB
Dominant pole	-7.88371×10^5 Hz
Second pole	-3.14795×10^9 Hz
Third pole	-9.24943×10^9 Hz
Zero 1	-3.14453×10^9 Hz
Zero 2	$+7.42947 \times 10^{10}$ Hz
Approx. 3-dB bandwidth from dominant pole	788.371 kHz
Unity-gain frequency	37.0662 MHz

5 Gain Margin and Phase Margin Discussion

The supplied Cadence frequency-response plots are forward amplifier transfer functions, not complete feedback-loop gain plots. The directly supported quantities are the 3-dB bandwidth estimates and the 0 dB crossing (unity-gain frequency). A formal gain margin requires a phase-crossover frequency where the loop phase reaches -180° , while a formal phase margin requires the loop phase at the gain-crossover frequency.

Accordingly, no numerical GM or PM values are assigned in this report without a defined feedback-loop transfer function. The reported 0 dB crossings are used only as the unity-gain frequencies of the displayed forward amplifier responses.

Table 6: Directly measurable frequency-domain quantities from the submitted forward-gain plots

Quantity	Part A	Part B
Low-frequency gain	18.3546 dB	33.1832 dB
Approx. 3-dB bandwidth	20.09 MHz	788.371 kHz
Unity-gain frequency	169.93 MHz	37.0662 MHz
Formal GM / PM	Not assigned	Not assigned

6 Comparison and Conclusion

6.1 Comparison of Part A and Part B

Table 7: Comparison of the two implemented cascode amplifiers

Parameter	Part A: Resistive Load	Part B: PMOS Cascode Current-Source Load
Load type	1.6 k Ω resistor	PMOS cascode current-source load
V_{DD}	1.8 V	1.8 V
C_L	5 pF	5 pF
NMOS widths	30, 30 μm	6.469202, 9.433895 μm
PMOS widths	-	20.415090, 20.415090 μm
PZ DC gain	8.27430 V/V	45.6276 V/V
Low-frequency gain	18.3546 dB	33.1832 dB
Approx. 3-dB bandwidth	20.09 MHz	788.371 kHz
Unity-gain frequency	169.93 MHz	37.0662 MHz

The two circuits illustrate the expected trade-off between gain and bandwidth for high-output-resistance loading. The resistive-load version has a lower extracted gain and a much higher dominant-pole frequency. The PMOS cascode current-source version provides a much higher gain because the active load increases the effective output resistance, while its dominant pole is at a substantially lower frequency.

6.2 Observations

Part A

1. The starting design calculations give $R_D \approx 1.6 \text{ k}\Omega$, $I_D \approx 555.6 \text{ }\mu\text{A}$ and a design g_m of approximately 7.5 mS.
2. The handwritten sizing calculations use $I_D = 0.56 \text{ mA}$ with $I_D/W = 18.60705$ and $18.8739 \text{ }\mu\text{A}/\mu\text{m}$, giving approximately 30 μm for both NMOS devices.
3. The PZ analysis reports a DC gain of 8.27430 V/V and a dominant pole near 20.09 MHz.

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4. The final dB plot gives 18.3546 dB low-frequency gain and a 169.93 MHz 0 dB crossing.

Part B

1. The four individual width sweeps select 6.469202 μm and 9.433895 μm for the lower and upper NMOS devices.
2. The lower and upper PMOS width sweeps both select 20.415090 μm at a drain-current magnitude of 100 μA .
3. The final PZ result gives a DC gain of 45.6276 V/V, with a dominant pole near 788.371 kHz.
4. The final dB plot gives 33.1832 dB low-frequency gain and a 37.0662 MHz 0 dB crossing.

6.3 Conclusion

A CMOS cascode amplifier was designed and analysed in Cadence Virtuoso for two loading configurations. In Part A, the resistive-load cascode uses a 1.6 k Ω resistor and two 30 μm NMOS transistors, as obtained from the handwritten sizing calculations. The submitted pole-zero and frequency-response results give a DC gain of 8.27430 V/V, a low-frequency gain of 18.3546 dB, an approximately 20.09 MHz dominant-pole 3-dB bandwidth and a 169.93 MHz unity-gain frequency.

In Part B, the resistor is replaced by a PMOS cascode current-source load. The four transistor widths are obtained from the supplied 100 μA width sweeps, giving 6.469202 μm and 9.433895 μm for the NMOS devices and 20.415090 μm for each PMOS device. The final response gives a PZ DC gain of 45.6276 V/V, a low-frequency gain of 33.1832 dB, an approximately 788.371 kHz dominant-pole 3-dB bandwidth and a 37.0662 MHz unity-gain frequency.

The simulations therefore demonstrate the effect of replacing a resistive load with a cascode current-source load: the active-load topology increases the effective output resistance and the voltage gain, while the dominant-pole frequency is reduced in the submitted implementation. Formal gain-margin and phase-margin values are not assigned because the supplied data are forward amplifier transfer functions rather than a complete feedback-loop transfer function.